

Problems Our Proposed Solutions

Limited Access to Eye Specialists Rural patients often need to travel to distant hospitals for specialist screening, increasing the time and cost of accessing care. **Screening at Local Health Centres** Enable trained healthcare workers to capture retinal images and obtain AI-assisted screening results, with remote specialist review.

Delayed Detection of Eye Damage Early diabetic eye damage can go unnoticed, delaying screening and timely referral. **Early Detection and Referral Support** Analyse retinal images for disease signs, estimate DR severity, and flag potentially referable cases for clinician assessment.

Time-Consuming Manual Screening Reviewing large numbers of retinal images places a heavy workload on limited specialists and can delay assessment. **Automated Analysis with Focused Review** Generate preliminary grades and annotated reports, helping specialists focus on flagged findings and uncertain cases.

Poor-Quality Retinal Images Blurry, unevenly lit, or incomplete images make reliable assessment difficult and may require repeat capture. **Smart Image Quality Checks** Check image quality before grading, enhance suitable borderline images, and provide specific recapture instructions for unusable images.

AI Results Without Clear Evidence A disease label or confidence score alone does not show clinicians which findings support the prediction. **Explainable, Evidence-Backed Results** Display AI attention maps, suspected lesion locations, and a short evidence summary. Flag uncertainty or conflicting findings for review.

Connectivity and Review Bottlenecks Weak internet connectivity and limited review capacity can interrupt the journey from image capture to specialist assessment. **Resumable Uploads and Capacity Planning** Preserve pending uploads and track review status. Use Simulink to assess bandwidth, processing, and staffing needs before scaling.